Memory Forensics Framework MFF v2

Cross-Case Comparison Report

Generated	2026-04-18 19:45:58 UTC
Cases compared	4
Case labels	Baseline Vs Case02 T1055 5 Attack · Baseline Vs Case03 T1059 Attack · Baseline Vs Case04 T157

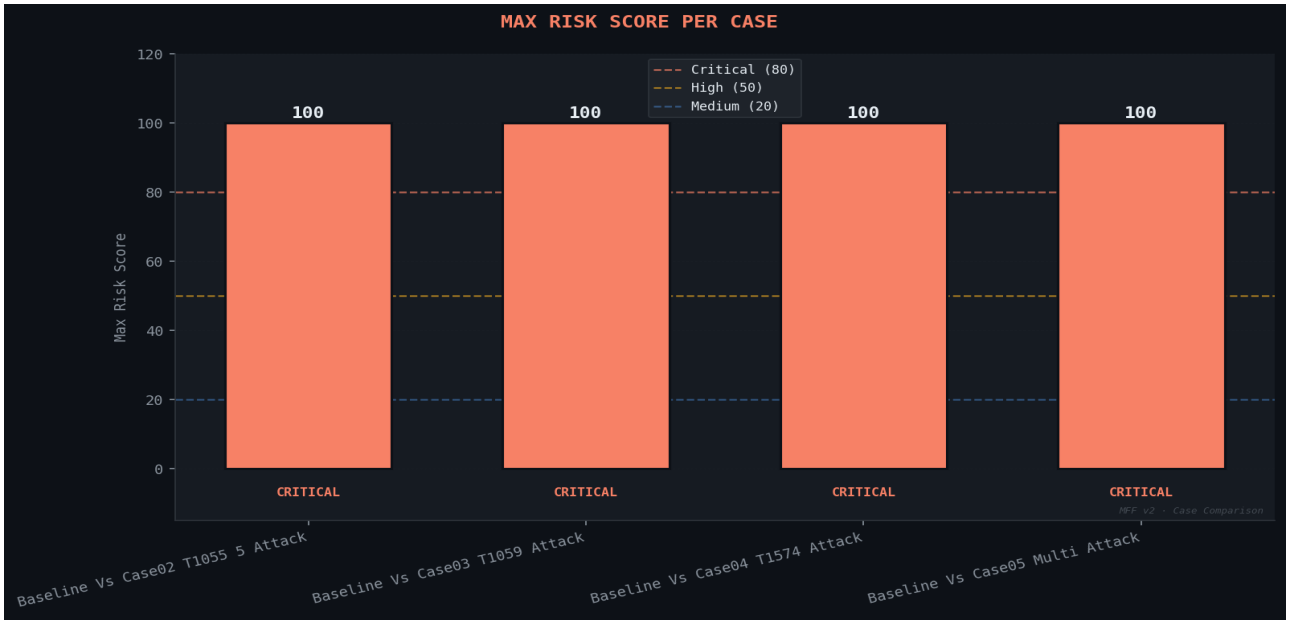
1. EXECUTIVE SUMMARY

Case	Severity	Max Score	Critical	Techniques	RWX	DLL Hijack	IOCs	New Procs
Baseline Vs Case02 T1055 5 Attack	CRITICAL	100	2	2	7	2	40	13
Baseline Vs Case03 T1059 Attack	CRITICAL	100	3	2	5	1	52	16
Baseline Vs Case04 T1574 Attack	CRITICAL	100	6	3	22	9	49	15
Baseline Vs Case05 Multi Attack	CRITICAL	100	8	8	42	12	59	16

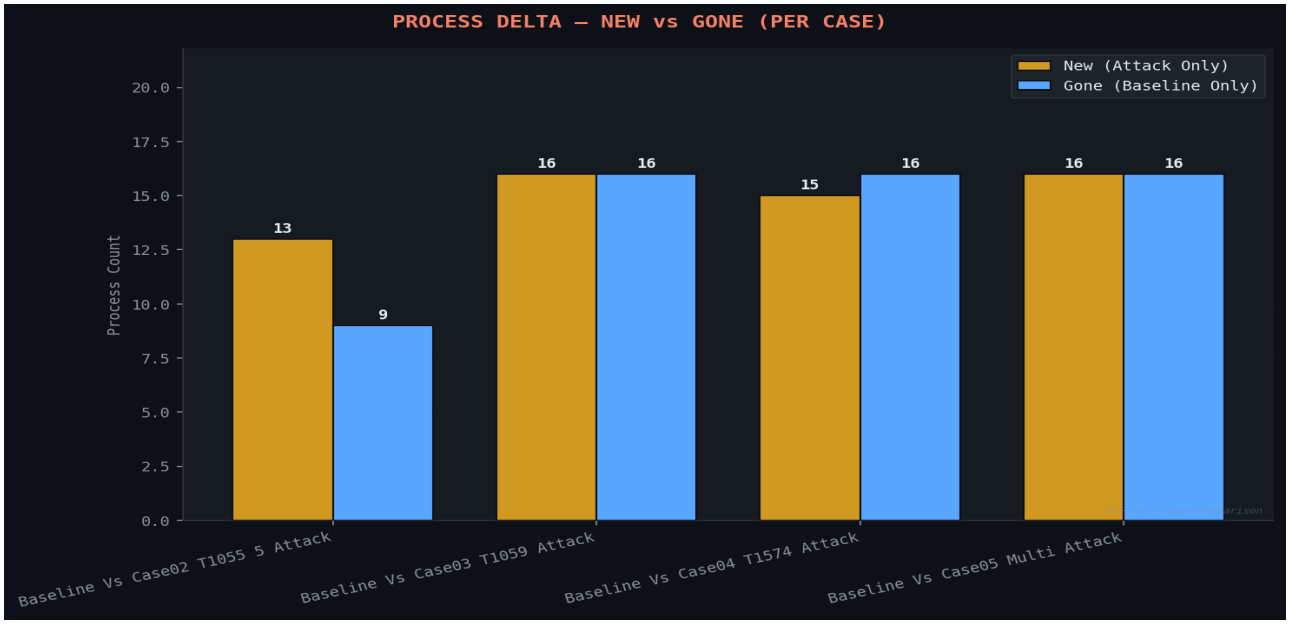
2. COMPARISON DASHBOARD

Case	Severity	Max Score	Critical	Techniques	RWX	DLL Hijack	IOCs	New Procs
Baseline Vs Case02 T1055 5 Attac	CRITICAL	100	2	2	7	2	40	13
Baseline Vs Case03 T1059 Attack	CRITICAL	100	3	2	5	1	52	16
Baseline Vs Case04 T1574 Attack	CRITICAL	100	6	3	22	9	49	15
Baseline Vs Case05 Multi Attack	CRITICAL	100	8	8	42	12	59	16

3. RISK SCORE COMPARISON

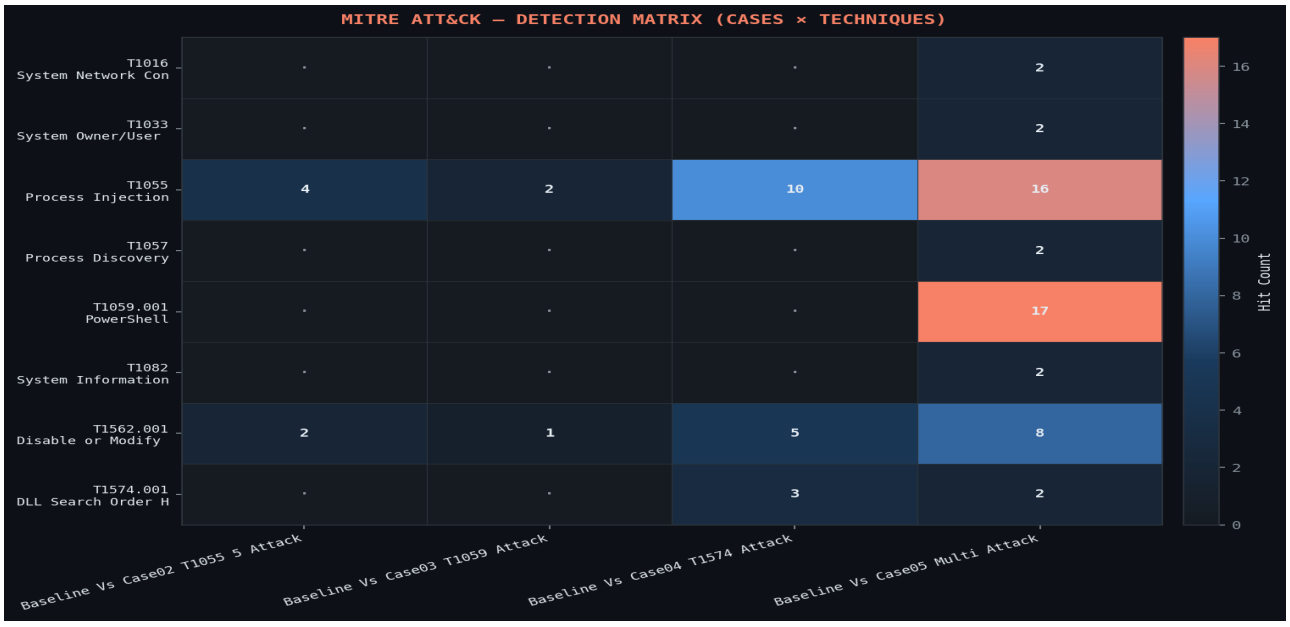


4. PROCESS DELTA

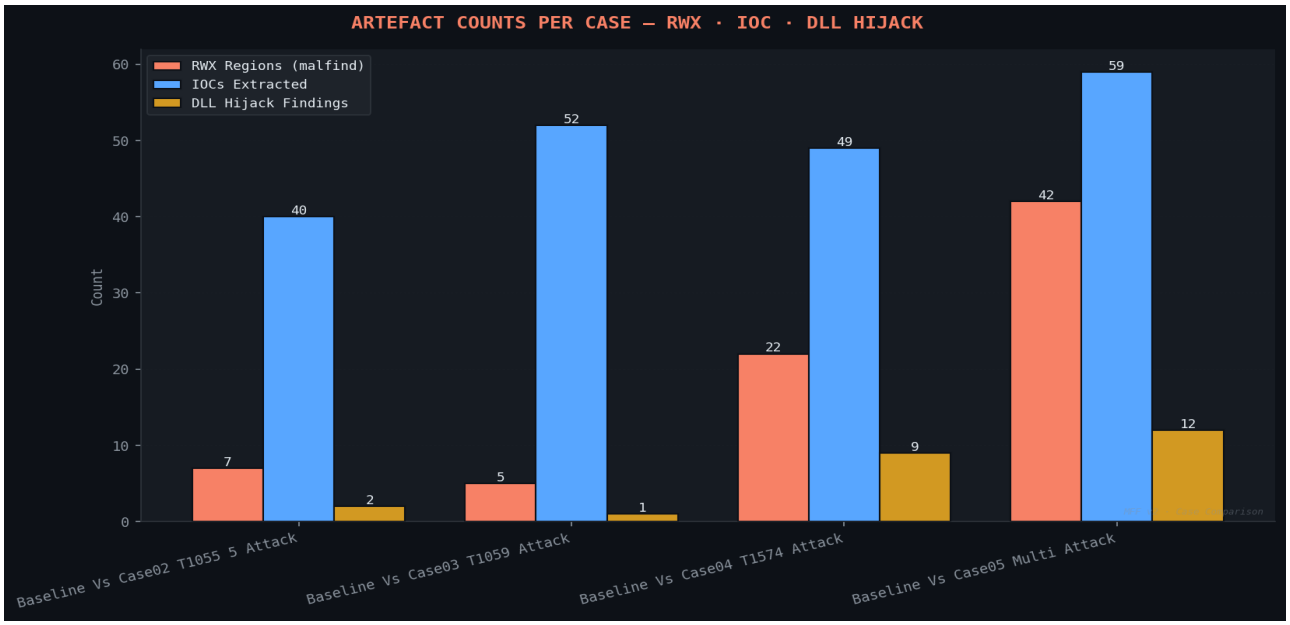


5. ATT&CK; TACTIC COVERAGE

6. MITRE ATT&CK; MATRIX



7. ARTEFACT COUNTS


ARTEFACT COUNTS PER CASE — RWX · IOC · DLL HIJACK
RWX Regions (malfind)
IOCs Extracted
DLL Hijack Findings
Count
0
10
20
30
40
50
60
7
40
2
5
52
1
22
49
9
42
59
12
Baseline Vs Case02 T1055 5 Attack
Baseline Vs Case03 T1059 Attack
Baseline Vs Case04 T1574 Attack
Baseline Vs Case05 Multi Attack

8. TECHNIQUE DETECTION MATRIX

Technique	Tactic	Name	Baseline Vs Ca	Baseline Vs Ca	Baseline Vs Ca	Baseline Vs Ca
T1016	Discovery	System Network Configuration Discovery	·	·	·	✓
T1033	Discovery	System Owner/User Discovery	·	·	·	✓
T1055	Defense Evasion	Process Injection	✓	✓	✓	✓
T1057	Discovery	Process Discovery	·	·	·	✓
T1059.001	Execution	PowerShell	·	·	·	✓
T1082	Discovery	System Information Discovery	·	·	·	✓
T1562.001	Defense Evasion	Disable or Modify Tools - AMSI Memory Patch	✓	✓	✓	✓
T1574.001	Defense Evasion	DLL Search Order Hijacking	·	·	✓	✓

9. INDIVIDUAL CASE SUMMARIES

Baseline Vs Case02 T1055 5 Attack

Severity	CRITICAL
Highest risk process	powershell.exe (score: 100)
Techniques detected	T1055, T1562.001
New processes	13
RWX memory regions	7
DLL hijack findings	2
IOCs extracted	40
Net new connections	131

Baseline Vs Case03 T1059 Attack

Severity	CRITICAL
Highest risk process	powershell.exe (score: 100)
Techniques detected	T1055, T1562.001
New processes	16
RWX memory regions	5
DLL hijack findings	1
IOCs extracted	52
Net new connections	133

Baseline Vs Case04 T1574 Attack

Severity	CRITICAL
Highest risk process	powershell.exe (score: 100)
Techniques detected	T1055, T1562.001, T1574.001
New processes	15
RWX memory regions	22
DLL hijack findings	9
IOCs extracted	49
Net new connections	94

Baseline Vs Case05 Multi Attack

Severity	CRITICAL
Highest risk process	powershell.exe (score: 100)
Techniques detected	T1016, T1033, T1055, T1057, T1059.001, T1082, T1562.001, T1574.001
New processes	16
RWX memory regions	42
DLL hijack findings	12
IOCs extracted	59
Net new connections	139

10. METHODOLOGY

Comparison method: Each attack case analysed independently against the same baseline using comparison_engine_v2.py. Results (CSV + JSON) loaded by this comparison module and aggregated into cross-case charts and tables.

Risk scoring: Rule-based transparent scoring (no ML). RWX +60 · Suspicious cmdline +40 · Malicious name +50 · DLL hijack +30 · Anomaly bonus +15.

DLL analysis: dll_analysis.py checks windows.dlllist CSV for protected DLLs (amsi.dll, version.dll, etc.) loaded from non-System32 paths.

ATT&CK; mapping: mitre_tagger.py — 30+ rule-based signatures across 8 tactics. Includes T1574.001 DLL hijacking, T1562 defense evasion, T1059 execution.

Framework: MFF v2 — Python Post-Volatility Memory Forensics Framework. FYP — University of Roehampton 2026. Modules: comparison_engine_v2, dll_analysis, mitre_tagger, network_ioc, process_tree, case_comparison, report_generator.